Note on: *Constant Maturity Asset Swap Convexity Correction*

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Reading notes

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One-paragraph summary

The paper prices an "ASW-let" — a payoff that pays a fixing of the forward asset swap spread of a reference bond at a future date — as the constant-maturity analogue of a CMS-let. The forward ASW spread R_t^{asw} is shown to be a ratio of risk-free tradables and so a martingale under the swaption measure A_t. Under the Hunt–Kennedy/Pelsser one-factor linear swap-rate model ($D_{T_0U}/A_{T_0} = \alpha + \beta_U R_{T_0}^{swp}$), the convexity correction reduces to a single covariance $\mathbb{E}_0^N[R_{T_0}^{asw} R_{T_0}^{swp}]$ between the ASW spread and the swap rate. With both rates modelled as displaced-lognormal (allowing negative ASW spreads — relevant for premium bonds), the correction admits a clean closed form. A numerical example calibrated to plausible Italian sovereign levels (ASW = 490 bp, swap = 4.29%, 5Y) gives convexity corrections in the range -0.19% to $+0.37\%$ depending on correlation $\rho \in \{-90\%, \dots, +90\%\}$ and ASW vol $\sigma^{asw} \in \{20\%, 30\%, 50\%\}$.

1 Setup and notation

Paper	House	Meaning
$R^{asw}_{tT_0}$	R^{asw}_t	Forward asset swap spread at t for ASW starting at T_0
R^{swp}_t	R^{swp}_t	Forward swap rate at t, same tenor as the ASW
B_t	$P_t(\cdot)$	Reference bond price process
$B^{\text{risk-free}}_t$	—	Hypothetical risk-free version of the same bond (benchmark)
D_{tT}	$D_{t,T}$	Risk-free discount factor
A_t	A_t	Risk-free annuity, $\sum_i \alpha_i D_{t,T_i}$
N_t	N_t	Numeraire (paper picks $N = A$)
T_0, T_p	T_0, T_p	Fixing date, payment date ($T_p \geq T_0$, usually $T_p = T_1$)
α, β_U	α, β_U	Linear-model coefficients
$\sigma^{swp}, \sigma^{asw}$	—	Displaced-lognormal vols
a^{swp}, a^{asw}	—	Displacements
ρ	ρ	Instantaneous correlation between the two driving BMs
CC	—	Convexity correction (definition in eq. 2)

2 Key equations

Forward ASW spread (definition):

$$R^{asw}_{tT_0} \;\equiv\; \frac{B^{\text{risk-free}}_t - B_t}{A_t}. \tag{1}$$

Says: the forward ASW spread is the gap between a risk-free benchmark bond and the actual bond, expressed as a per-period annuity-weighted rate. As a ratio of tradable assets, it is a martingale under the swaption measure (with numeraire A).

Convexity correction (definition):

$$CC \;\equiv\; \frac{N_0}{D_{0,T_p}}\,\mathbb{E}^N_0\left[R^{asw}_{T_0}\,D_{T_0,T_p}/N_{T_0}\right] \;-\; R^{asw}_0. \tag{2}$$

Says: the correction is the difference between the forward price of an $R^{asw}_{T_0}$ payment at T_p and the “no-volatility” value $R^{asw}_0 D_{0,T_p}$.

Linear swap-rate model.

$$\frac{D_{T_0,U}}{A_{T_0}} \;=\; G_U(R^{swp}_{T_0}) \;=\; \alpha + \beta_U R^{swp}_{T_0}, \tag{3}$$

for all $U \in \{T_p, T_1, \ldots, T_n\}$, with

$$\alpha \;=\; \frac{1}{\sum_{i=1}^n \alpha_i}, \qquad \beta_U \;=\; \frac{D_{0,U}/A_0 - \alpha}{R^{swp}_0}. \tag{4}$$

Says: same calibration as in Pucci’s CMT paper; here the swap rate plays the role that R^{cmt} played there.

Linear-swap ASW convexity correction (Proposition 1):

$$CC \;=\; \left(1-\frac{A_0\,\alpha}{D_{0,T_p}}\right)\left(\frac{\mathbb{E}_0^N[R^{asw}_{T_0}R^{swp}_{T_0}]}{R^{swp}_0}-R^{asw}_0\right). \tag{5}$$

Says: the correction is the product of (i) a discount-curve geometric factor and (ii) the covariance excess of $R^{asw}_{T_0}R^{swp}_{T_0}$ over its deterministic value. Sign follows the sign of correlation when both rates are martingales.

Displaced-lognormal dynamics:

$$\frac{\mathrm{d}R^{swp}_t}{R^{swp}_t+a^{swp}} = \sigma^{swp}\;\mathrm{d}W^{swp}_t, \tag{6}$$

$$\frac{\mathrm{d}R^{asw}_t}{R^{asw}_t+a^{asw}} = \sigma^{asw}\;\mathrm{d}W^{asw}_t, \tag{7}$$

$$\rho = \mathrm{d}\langle W^{swp},W^{asw}\rangle_t/\,\mathrm{d}t. \tag{8}$$

Says: the displacement a allows the rate to become negative; this matters here because asset-swap spreads can be negative on premium bonds (mentioned in the introduction).

Cross-expectation under displaced-lognormal (eq. (7) in body):

$$\begin{aligned}\mathbb{E}_0^N[R^{swp}_{T_0}R^{asw}_{T_0}] \;&= (R^{swp}_0+a^{swp})(R^{asw}_0+a^{asw})\;\exp(\sigma^{swp}\sigma^{asw}\,\rho\,T_0)\\ &\quad - a^{swp}(R^{asw}_0+a^{asw})-a^{asw}(R^{swp}_0+a^{swp})+a^{swp}a^{asw}.\end{aligned} \tag{9}$$

Says: closed-form cross-moment in the DL model. Substitute into (5) for the full CC.

Pure lognormal limit (Corollary 3, $a^{swp}=a^{asw}=0$):

$$CC \;=\; R^{asw}_0\left(1-\frac{A_0\,\alpha}{D_{0,T_p}}\right)(\exp(\sigma^{swp}\,\sigma^{asw}\,\rho\,T_0)-1). \tag{10}$$

Says: the cleanest case. The convexity correction is proportional to R^{asw}_0 and to the correlation-weighted covariance factor; sign of CC equals sign of ρ (with the geometric prefactor positive in the usual regime $A_0\,\alpha < D_{0,T_p}$).

Validity conditions for displaced-diffusion (Remark 2):

$$R^{asw}_0 > -a^{asw}\quad\text{(DL lognormal)}\qquad\text{or}\qquad 0<R^{asw}_0<-a^{asw}\quad\text{(DL anti-lognormal)}, \tag{11}$$

and similarly for R^{swp}.

3 Derivations & commentary

Commentary

The headline move is the same as in the CMT paper: identify R^{asw} with a ratio of tradables (here just on the risk-free curve, no defaultable processes needed). Under the swaption measure with numeraire A, R^{asw} is a martingale and CMS machinery transfers directly. The catch is that the dynamics of R^{asw} and R^{swp} must be modelled jointly — CC depends on their *cross-moment*, not just on individual variances. So this is a correlation product, with all the calibration problems that implies.

Commentary

The choice of displaced-lognormal instead of plain lognormal is operationally important: ASW spreads on premium bonds (where the bond's coupon exceeds market levels) can be *negative*. The paper notes that this used to be the case for treasuries when interbank deposits were perceived less risk-free. A pure lognormal process forbids this, so DL is the safer modelling default even when $R_0^{asw} > 0$ today.

Commentary

The paper's §8 honestly enumerates the model's weaknesses: **(i)** the linear-G_U assumption may not be realistic for long-tenor bonds (richer functional forms in Hagan's Convexity Conundrums exist); **(ii)** the ASW vol is hard to imply since liquid ASW options don't exist — historical estimation is needed, possibly supplemented by bond-option quotes; **(iii)** static replication a la Hagan doesn't trivially apply to ASW-let because the swaption-measure formula contains *both* $R_{T_0}^{asw}$ and $R_{T_0}^{swp}$ explicitly; **(iv)** correlation is also not market-implied.

Commentary

The forward ASW package embeds a forward position on the underlying bond. Forward repo on sovereigns is thin; the paper notes that for issuers with a deep curve (FR, IT, BE, DE) the forward bond can be synthesised by coupon-stripping or bootstrapping a zero curve.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

Quantities the paper defines:

- Forward ASW spread $R_{tT_0}^{asw}$ via (1).
- Convexity correction CC via (2).
- Linear-model coefficients α, β_U via (4).
- Linear-swap closed-form CC via (5).
- Cross-expectation under DL dynamics (9), and its lognormal limit (10).

Canonical numerical case (paper Tables 1, 4, 5; section 7):

- Reference: Italian sovereign bond schedule (paper Tab. 4); $T_p = 08/02/2017$ (one specific period).
- Forward swap rate $R_0^{swp} = 4.29\%$ (matched to discount curve, Tab. 5).
- Forward ASW spread $R_0^{asw} = 490$ bp (sovereign-flavoured).
- Forward swap-rate vol $\sigma^{swp} = 30\%$.
- Tested grid: $\sigma^{asw} \in \{20\%, 30\%, 50\%\}$, $\rho \in \{-90\%, -50\%, 0\%, +50\%, +90\%\}$.

Expected numerical values (Table 2 of the paper, pure-lognormal CC):

$\sigma^{asw} \backslash \rho$	-90%	-50%	0%	$+50\%$	$+90\%$
20%	-0.09%	-0.05%	0.00%	$+0.06\%$	$+0.12\%$
30%	-0.13%	-0.08%	0.00%	$+0.10\%$	$+0.19\%$
50%	-0.19%	-0.12%	0.00%	$+0.18\%$	$+0.37\%$

Sensitivity to ASW slope (paper Table 3, DL skew impact):

$\sigma^{asw}\backslash\rho$	-90%	-50%	0%	$+50\%$	$+90\%$
20%	0.02%	0.01%	0.00%	0.01%	0.03%
30%	0.05%	0.02%	0.00%	0.02%	0.08%
50%	0.11%	0.04%	0.00%	0.07%	0.28%

Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(1)	$R^{asw}_{tT_0}$	forward ASW spread as ratio of risk-free tradables
(2)	CC	convexity correction (definition)
(3)–(4)	α, β_U	linear swap-rate model coefficients
(5)	CC	closed-form CC under linear model
(9)	$\mathbb{E}^N_0[R^{swp}_{T_0} R^{asw}_{T_0}]$	DL cross-expectation
(10)	CC (lognormal limit)	pure lognormal closed form

Invariants and identities to verify:

1. *Reproduction of Table 2*: the lognormal closed form (10) with $R^{asw}_0 = 490$ bp, $\sigma^{swp} = 30\%$, $T_0 \approx 5$Y, and the appropriate $A_0\alpha/D_{0,T_p}$ factor should reproduce the $\sigma^{asw} \times \rho$ grid above to within 0.01%.
2. *Linear-calibration consistency*: $\alpha + \beta_{T_p} R^{swp}_0 = D_{0,T_p}/A_0$ and the sum $\sum_i \alpha_i(\alpha + \beta_{T_i} R^{swp}_0) = 1$ hold by construction.
3. *DL $\to$ lognormal limit*: as $a^{swp}, a^{asw} \to 0$, (9) reduces to $(R^{swp}_0)(R^{asw}_0)\exp(\sigma^{swp}\sigma^{asw}\rho T_0)$, and the full CC reduces to (10).
4. *Sign of CC*: under lognormal dynamics and the typical regime $A_0\alpha < D_{0,T_p}$, the sign of CC equals the sign of ρ.
5. *Vanishing at zero vol*: $CC = 0$ whenever $\sigma^{swp}\sigma^{asw} = 0$ (lognormal limit), or whenever both rates are independent ($\rho = 0$ in lognormal).
6. *DL validity*: the conditions (11) must hold for the chosen (a^{swp}, a^{asw}) pair; otherwise the implied dynamics are inadmissible.

5 Glossary of symbols (paper's notation)

$R^{asw}_{tT_0}$	Forward asset swap spread at t for ASW starting at T_0
R^{swp}_t	Forward swap rate at t
B_t	Reference bond price process
$B^{\text{risk-free}}_t$	Hypothetical risk-free version of the same bond
D_{tT}	Risk-free DF
A_t	Risk-free annuity over the schedule
T_0	Fixing date
T_p	Payment date, $T_p \geq T_0$
$T_1, \ldots, T_n$	Bond / annuity schedule dates
$yf(T_{i-1}, T_i)$	Year fraction (often denoted α_i in the paper)
N_t	Numeraire process (paper picks $N = A$)
α, β_U	Linear-model coefficients
$\sigma^{swp}, \sigma^{asw}$	DL vols of the swap rate / ASW spread
a^{swp}, a^{asw}	DL displacements
ρ	Instantaneous correlation
W^{swp}, W^{asw}	Driving Brownian motions
CC	Convexity correction (paper's eq. (3))

TODO / Open questions

- Reproduce Table 2 in full (lognormal grid) and Table 3 in full (DL slope sensitivity) using (10) and (9).
- Schedule used in the paper's example: bond schedule from Tab. 4 (not transcribed here); recover from paper if a full end-to-end reproduction is needed.
- Discount curve from Tab. 5: not transcribed; the linear-model coefficients α, β_U for the canonical case must be derived from the same curve to reproduce Table 2.
- Multi-curve generalisation: the paper notes (end of §5) that formula (5) extends naturally to a multi-curve framework where the swap-rate projection and discount are independent. Not in scope for first-pass validation but worth bookmarking.